

PLANETARY PHOTOMETRY



Planetary Photometry
University of Victoria

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University of Victoria

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Licensing

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About this Book

Max Fairbairn, who was my first ever graduate student, obtained an M.Sc. in Astronomy at the University of Victoria, Canada, in 1971, after which he returned to his native Australia. It was some years later when I heard from him again. Apart from wanting to renew a friendship, he had found himself interested in a paper that I and two other Victoria students, Tim Lester and Marshall McCall, had written on the subject of planetary photometry. This led to some fruitful correspondence on the subject, during which it became evident to me that Max understood the subject far better than I had ever done. This is, of course, the greatest compliment that a former student can ever pay to his erstwhile professor. His email letters to me on the subject were so clear that I suggested that it would be useful to put his material on the Web so that others could benefit from the thought he had put into it.

Max had just started to put his material together when he died, after a painful illness, in January 2005, at the young age of 58. At that time, he had put one chapter, "Principles of Planetary Photometry", on the Web, though there were some problems with the diagrams, which had not reproduced well on the Web, and he had written, but not posted, a second chapter, "Albedo". He had planned further chapters, leading up to the determination of asteroid shapes from their lightcurves, but was not able to complete these before his death.

Nevertheless, although this work is far from complete, I felt that the first two chapters, which deal with fundamental principles and definitions, described in a clear and unambiguous manner, would be of interest and use not only to students who are just starting out on the subject, but quite likely also to more experienced astronomers who are actively working in the field. I have therefore repaired the computer-mangled defective drawings that appeared on the Web posting of Chapter 1 and have done some minor editing, and have copy-edited the unpublished Chapter 2, and I have called upon the computer expertise of Jason Stumpf to post these two chapters. We also found a document entitled *A Brief History of the Lommel-Seeliger Law*, which I have treated as a third chapter. Although there is much more to the subject, I am hopeful that some readers will find this introductory material of interest and use, and that Max's efforts will then not have been in vain.

Comments and notification of errors can be directed to the undersigned at jtatum@uvic.ca

CHAPTER OVERVIEW

1: Principles of Planetary Photometry

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- 1.2: Radiance and the Equation of Transfer
- 1.3: Diffuse Reflection and Transmission
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Thumbnail: Surface of Mercury (Public Domain; NASA).

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